Miraslau Pisaryk

Java / Spring Boot Backend Engineer | Commercial Experience

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github.com/CHIKOJgg | linkedin.com/in/miroslav-pisaryk-953490261 | **Open to Remote Work (Worldwide / B2B)**

SUMMARY

Backend engineer with commercial experience designing and maintaining high-reliability server-side applications on Java 21/25 and Spring Boot 3/4. Focused on transactional systems (FinTech), real-time data streaming (WebSocket STOMP, Redis), event-driven microservices (RabbitMQ, Multi-Module Monorepo), and AI agent integrations with Spring AI, Keycloak 26, and PostgreSQL 17 + pgvector RAG. Agile (Scrum) practitioner; ensuring 100% integration test isolation via Testcontainers. English C1 Advanced.

SKILLS

Languages: Java (17, 21, 25), SQL, TypeScript/JavaScript (Foundations)

Java Core: Collections, OOP, Generics, Stream API, Concurrency (Virtual Threads / Loom, CompletableFuture, ConcurrentHashMap)

Frameworks: Spring Boot 3/4, Spring AI, Spring Cloud OpenFeign, Spring Data JPA / Hibernate, Spring Security, WebSocket (STOMP), Resilience4j, Bucket4j, MapStruct

IAM & Security: Keycloak 26 (OAuth2 Resource Server, OIDC, PKCE, JWT signature verification via JWKS, JIT Provisioning, TokenRelay)

Messaging: RabbitMQ 3.13 (Topic Exchanges, JSON serialization, Dead-Letter Queues, Decoupled Domain Events)

Databases: PostgreSQL 17, pgvector (Vector Embeddings, RAG), Redis 7 (Caching, Pub/Sub, Rate Limiting), Flyway, EXPLAIN ANALYZE

DevOps & Testing: Docker, Docker Compose, GitHub Actions (CI/CD), Maven, Linux/Bash, JUnit 5, AssertJ, Mockito, Testcontainers

Architecture: Distributed Microservices, Multi-Module Monorepo, Event-Driven Architecture, DDD, Clean Architecture, REST RFC 7807

Spoken Languages: English C1 Advanced (EF SET 70/100, verified); Polish — beginner

WORK EXPERIENCE

Nexgen Soft — Java Backend Developer

Java 21, Spring Boot 3.3, PostgreSQL, Redis, WebSocket (STOMP), JWT, Flyway, Docker, GitHub Actions
September 2025 – Present | Remote | Outsourcing company, FinTech distributed team

- **FinTech Platform:** Developing the server side of a FinTech platform for account management, trading operations, and real-time quotes in a distributed team of 8.
- **Matching Engine:** Designed and implemented a thread-safe in-memory order matching engine (price-time priority) for limit and market orders using Java Concurrency primitives.
- **Redis Optimization:** Integrated Redis for order book caching and message queuing, reducing PostgreSQL load by ~40% during peak hours.
- **WebSocket Streaming:** Engineered real-time order book and notification streaming over WebSocket (STOMP) for 200+ concurrent clients with sub-second latency.
- **Schema & Migrations:** Architected a 13-table PostgreSQL schema with double-entry accounting ledgers and audit trails; automated migrations via Flyway.
- **Security & CI/CD:** Implemented Spring Security with JWT token rotation; maintained test coverage via JUnit 5 and Testcontainers; set up CI/CD via GitHub Actions.

DataBridge Solutions — Software Engineer (Distributed Microservices & AI Integration)

Java 25 (Virtual Threads), Spring Boot 3.4, Spring AI, PostgreSQL 17 + pgvector, Keycloak 26, RabbitMQ 3, OpenFeign, Redis 7, Testcontainers
April 2025 – September 2025 | Remote | Multi-Module Maven Microservices Monorepo

- **Microservices Monorepo:** Architected and executed the migration from a monolith to an Event-Driven Microservices Architecture structured as a Maven multi-module monorepo (klawa-contracts, klawa-core, klawa-ai-agent).

- **Core & Virtual Threads:** Developed the high-throughput klawa-core service on Java 25 utilizing Virtual Threads (Loom) for non-blocking I/O across task orchestration and scheduling.
- **Keycloak 26 IAM:** Integrated Keycloak 26 as an OAuth2 Resource Server; implemented stateless JWT signature verification via JWKS, realm role mapping, and Just-In-Time (JIT) user provisioning.
- **Spring AI & pgvector RAG:** Engineered the klawa-ai-agent microservice on Spring AI; implemented Retrieval-Augmented Generation (RAG) using PostgreSQL 17 with pgvector for real-time semantic similarity search.
- **Function Calling (@Tool):** Implemented @Tool capabilities allowing the LLM to autonomously trigger CRUD commands in klawa-core via Spring Cloud OpenFeign with TokenRelayInterceptor (JWT propagation).
- **RabbitMQ Broker:** Configured RabbitMQ 3.13 (Topic Exchange klawa.events) for asynchronous domain events (TaskCreatedEvent, AiTaskAnalysisRequestedEvent); decoupled vector embedding generation.
- **Resilience & Testcontainers:** Integrated Resilience4j Circuit Breakers, Bucket4j token bucket rate limiting, Redis caching, and isolated integration tests using Testcontainers (PostgreSQL, Redis, RabbitMQ).

ADDITIONAL PROJECTS

Crypto Portfolio Optimizer — Quantitative Analysis & Backtesting Engine

Java 21, JavaFX, ojalgo, CoinGecko API, Maven | github.com/CHIKOJgg/MarkowitzModel-java-crypto-analysis

- **Quantitative Models:** Engineered 12 alpha models (Momentum, EWMA, RSI, MACD, Bollinger) and 9 allocation frameworks (Markowitz, Risk Parity, HRP, CVaR, Black-Litterman).
- **Backtesting & Stress Testing:** Built walk-forward backtesting engine with Monte Carlo simulation (GBM) and Brinson attribution, calculating Sharpe, Sortino, Calmar, and VaR/CVaR 95% metrics.

EDUCATION & CREDENTIALS

Belarusian State University of Informatics and Radioelectronics (BSUIR)

Bachelor's Degree in Information Systems and Technologies | 2024 – 2028 | Minsk, Belarus

Verified English Certificate: EF SET — Score 70/100, C1 Advanced (Official verification: cert.efset.org/ru/cAsbkU)